

ZHEKAI (ZACK) JIN

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EMPLOYMENT EXPERIENCE

NVIDIA Corporation

Shanghai, China

Senior Software Engineer - ML & GT Generation

Mar. 2024 - Present

- Deployed a production pose-adjustment pipeline that fused keyframe image-feature correspondences in a global optimizer, improving ground-truth poses for autonomous-driving perception models.
- Designed attribute-aware correspondence tracking and dynamic covariance modeling for the optimizer, reducing production pose error by 75.6% and pose variation by 65.2%.
- Built an automated regression framework and clip-level Pose Accuracy dashboard to detect long-tail failures, benchmark algorithm changes, and guide production rollouts.
- Built a seven-camera multimodal workflow aligning video, 2D/3D labels, HD maps, ego trajectories, and VLM-generated captions; replaced Cosmos Predict 2.5 joint self-attention with factorized temporal and cross-view attention for memory-efficient long-horizon post-training.

Motional AD, Inc.

Pittsburgh, PA

Software Engineer II - Localization & Mapping

Jul. 2021 - Sep. 2023

- Rescued the localization system from critical edge-case failures by designing and productionizing a state-of-the-art feature-based downsampling algorithm robust to noise and outliers, while reducing scan-matching latency by 20%.
- Led a three-person team to develop an offline GTSAM pose-optimization pipeline for daily drivelog analysis, enabling accuracy benchmarking and regression validation of localization changes.
- Engineered a robust nonlinear step solver for Iterative Closest Point (ICP), improving scan-matching robustness and real-time localization performance in challenging scenarios.
- Unified online and offline health metrics with interactive visualizations, including custom localization health measures, enabling continuous monitoring and data-driven benchmarking.

SELECTED RELEVANT EXPERIENCE

Uber Advanced Technologies Group (ATG)

Pittsburgh, PA

Localization Engineering Intern - Mapping

Jun. 2020 - Aug. 2020

- Pioneered the industry's first objective metric for evaluating localized pose estimates independently of an HD map or ground truth, setting a new standard for localization evaluation in the field.
- Built a modular SLAM verification pipeline that achieved 98% classification accuracy on a custom dataset and supported additional pose-quality metrics.

Momenta.ai

Beijing, China

Research & Development Intern - LiDAR Perception

May 2018 - Aug. 2018

- Developed efficient ground-detection and semantic road-segmentation algorithms with 98% precision.
- Built a real-time object-tracking pipeline for sparse point clouds using 3D interpolation, deployed in Momenta's L4 autonomy stack.

EDUCATION

Carnegie Mellon University - School of Computer Science

Pittsburgh, PA

M.S. in Robotic Systems Development | GPA: 4.00

May 2021

The Cooper Union for the Advancement of Science and Art

New York, NY

B.Eng. in Electrical Engineering, Minor in Computer Science | GPA: 3.69

May 2019

SKILLS

Languages C++, Python, Java, C, SQL, Shell Scripting

Technology GTSAM, Ceres, PCL, OpenCV, ROS, PyTorch, Linux, CARLA, AWS EC2/S3